Reward Recovery in Sequential Decision Models: A Unified Treatment of Three Estimation Approaches

With Application to Lee, Sudhir & Wang (2026)

Pranjal Rawat*

April 2026 – Draft

*Georgetown University. pp712@georgetown.edu.

TL;DR. We compare three methods for recovering reward functions from observed sequential choices in dynamic discrete choice (DDC) models: (i) reduced-form Q-estimation (flexible logit, no Bellman equation), (ii) adversarial inverse reinforcement learning (AIRL, with and without anchor normalization), and (iii) inverse soft-Q learning (IQ-Learn). All three fit the observed policy equally well. They produce identical counterfactual predictions when the analyst changes only state values while holding the MDP structure fixed–for example, reducing wait times and simulating forward under the estimated policy. We call these Type I counterfactuals. The methods diverge when the counterfactual changes the environment itself: new transition dynamics, modified action sets, or segment-specific access rules. We call these Type II counterfactuals. Only methods that separate immediate reward from continuation value can re-solve the Bellman equation under the new environment. A reduced-form logit bundles the two together and fails on Type II with up to 18.5 percentage points of CCP error.

Executive takeaways:

1. Validation exercises do not discriminate between methods. The wait-time validation in Lee, Sudhir & Wang (2026, Section 6.3) is a Type I counterfactual. Every method that recovers the advantage function–including a simple reduced-form logit–passes this test identically. The AIRL machinery is not needed for it.
2. Policy counterfactuals require structural reward recovery. The WFF removal, segment-specific delay, and content-based paywall counterfactuals (Sections 7.1–7.3 of LSW) all change the MDP environment. These are Type II and require the structural reward separated from continuation values. A reduced-form model produces 18.5pp mean CCP error on the paywall counterfactual in our simulation.
3. Anchor normalization is the key identification device. Generic AIRL recovers reward only up to a potential-based shaping function, which is observationally equivalent in the original MDP but produces different counterfactual behavior when transitions change. The LSW anchors–exit payoff fixed to zero, absorbing-state value fixed to zero–eliminate this ambiguity exactly (Theorems 2–3), collapsing the identified set to a singleton.
4. The identification is robust, not knife-edge. Misspecifying the true exit payoff by 0.5 (against an anchor that assumes zero) produces only 0.2 percentage points of counterfactual error. The degradation is smooth and proportional.
5. Anchor constraints can be added to any estimator. IQ-Learn recovers the structural reward via the inverse Bellman operator without GAN training, but its χ^2 objective has multiple local optima in finite samples. A simpler approach–adding anchor penalties to the standard logit MLE–converges reliably and beats the unconstrained reduced-form at large N. The anchor principle (fixing the reward of a known action) is the transferable idea; the specific estimation framework (adversarial, ERM, penalized MLE) is secondary.

1 Environment

An agent faces a stationary, infinite-horizon, discrete-time decision problem. The primitives are a finite state space $\mathcal{S}$, a finite action space $\mathcal{A}$, a discount factor $\beta \in (0, 1)$, a Markov transition kernel $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, and a per-period reward function $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$, unknown to the econometrician.

The agent observes private additive shocks ε_a drawn i.i.d. across actions and periods from a Type I extreme value distribution (Rust, 1987). Per-period payoff from choosing a in state s is $r(s, a) + \varepsilon_a$. This framework is equivalent to maximum causal entropy IRL (Ziebart, 2010),

where the entropy regularizer replaces the Gumbel shocks. Both yield the same Bellman recursion, likelihood, and policy representation.

2 Bellman Recursion and the Optimal Policy

The *action-value function* (Q-function) satisfies

$$Q^*(s,a) = r(s,a) + \beta \sum_{s' \in \mathcal{S}} P(s' \mid s,a)\, V^*(s'), \tag{1}$$

where the *state-value function* is

$$V^*(s) = \log \sum_{a' \in \mathcal{A}} \exp\bigl(Q^*(s,a')\bigr). \tag{2}$$

The optimal policy takes the multinomial logit form:

$$\pi^*(a \mid s) = \frac{\exp\bigl(Q^*(s,a)\bigr)}{\sum_{a'} \exp\bigl(Q^*(s,a')\bigr)} = \exp\bigl(Q^*(s,a) - V^*(s)\bigr). \tag{3}$$

Define the *advantage function*:

$$A^*(s,a) \;=\; Q^*(s,a) - V^*(s) \;=\; \log \pi^*(a \mid s). \tag{4}$$

The advantage is identified directly from choice data: $\hat{A}(s,a) = \log \hat{\pi}(a \mid s)$. But A^* does not pin down r without further restrictions, because the Bellman equation bundles the reward with discounted future values.

3 What Is Identified from Choice Data Alone

From a panel of choices $\{(s_{it}, a_{it})\}$ and transitions, the econometrician identifies:

(i) Conditional choice probabilities $\hat{\pi}(a \mid s)$.

(ii) The transition kernel $\hat{P}(s' \mid s,a)$.

The Hotz–Miller (1993) inversion gives $Q^*(s,a) - Q^*(s,a_0) = \log \pi^*(a \mid s) - \log \pi^*(a_0 \mid s)$ for any baseline action a_0: Q-function differences across actions are identified, but levels are not. However, these Q-differences generally do not equal reward differences because they include continuation terms. Without additional structure or normalization that pins down levels (e.g., anchors), r is not identified in levels from choice data alone.

This is the Magnac and Thesmar (2002) non-identification result. Any $\tilde{r}(s,a) = r(s,a) + \lambda(s)$, where $\lambda : \mathcal{S} \to \mathbb{R}$ is action-independent, generates identical CCPs. Following Kalouptsidi et al. (2021), we call the set of all such $\tilde{r}$ the *identified set* of reward functions.

4 Approach 1: Reduced-Form Q-Estimation

This is the natural baseline: fit a flexible logit directly to the choice data, without imposing any dynamic structure.

Specification. Let $Q_\omega : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ be a neural network with parameters ω. Estimate by maximum likelihood:

$$\hat{\omega} = \arg\max_{\omega} \sum_{i,t} \log \pi_\omega(a_{it} \mid s_{it}), \qquad \pi_\omega(a \mid s) = \frac{\exp\big(Q_\omega(s,a)\big)}{\sum_{a'} \exp\big(Q_\omega(s,a')\big)}. \tag{5}$$

This is a standard multinomial logit with a flexible index. No Bellman equation is imposed. No transition data are used.

What it recovers. At the population optimum, $Q_{\hat{\omega}}(s,a) = Q^*(s,a) + c(s)$ for some action-independent $c : \mathcal{S} \to \mathbb{R}$, because the logit likelihood depends only on within-state action differences. The estimated policy matches the true policy exactly. The backed-out "reward" is

$$\hat{r}_{\mathrm{RF}}(s,a) \;\equiv\; \log \hat{\pi}(a \mid s) \;=\; A^*(s,a),$$

which is the advantage–not the true reward. It bundles immediate payoffs with continuation value differences.

Computational cost. Standard backpropagation. No bilevel optimization, no adversarial loop, no trajectory simulation.

5 Approach 2: Adversarial Inverse Reinforcement Learning

5.1 The Generic AIRL Framework (Fu et al., 2018)

AIRL alternates between two updates. A generator (policy π_ψ) produces trajectories by interacting with the MDP. A discriminator D_θ is trained via binary cross-entropy to distinguish expert transitions from generator transitions. The discriminator takes the structured form:

$$D_\theta(s,a,s') = \frac{\exp\big(f_\theta(s,a,s')\big)}{\exp\big(f_\theta(s,a,s')\big) + \pi_\psi(a \mid s)}, \tag{6}$$

where the policy probability $\pi_\psi(a \mid s)$ is "filled in" from the current generator (held fixed during discriminator updates). The key structural restriction is that f_θ decomposes as

$$f_\theta(s,a,s') = g_\theta(s,a) + \beta\, h_\phi(s') - h_\phi(s). \tag{7}$$

Here g_θ is a reward network and h_ϕ is a shaping network. The term $\beta h(s') - h(s)$ is a potential-based reward shaping (Ng et al., 1999) that does not alter the optimal policy.

The generator is updated using the shaped reward $\hat{r}(s,a) = \log D_\theta - \log(1 - D_\theta) = f_\theta(s,a,s') - \log \pi_\psi(a \mid s)$, which, when summed over trajectories, yields the entropy-regularized policy objective (Fu et al., 2018, Appendix A.3).

Convergence. At the global optimum of the GAN game–where the generator matches the expert, $\pi_\psi = \pi^*$–the discriminator outputs $D = 1/2$ everywhere. This requires $\exp\{f^*(s,a,s')\} = \pi^*(a \mid s)$, yielding (Fu et al., 2018, Appendix A.4):

$$f^*(s,a,s') = \log \pi^*(a \mid s) = A^*(s,a). \tag{8}$$

What the reward network g^* recovers. Substituting the decomposition (7) into (8) and using the Bellman equation $A^*(s,a) = r(s,a) + \beta \sum_{s'} P(s' \mid s,a)V^*(s') - V^*(s)$:

$$g^*(s,a) = r(s,a) - \delta(s) + \beta \sum_{s'} P(s' \mid s,a)\, \delta(s'), \tag{9}$$

where $\delta(s) \equiv V^*(s) - h^*(s)$ is the *shaping residual*. Unless $\delta \equiv 0$, the recovered reward g^* is a potential-based perturbation of the true reward r. Both generate the same optimal policy in the original MDP, but they can generate different counterfactual behavior when the transition kernel changes–this is precisely the "reward ambiguity problem" that Fu et al. develop in their Section 5.

The state-only restriction (Fu et al., 2018, Theorem 5.1 & Appendix C). If $g_\theta(s)$ depends only on the state (not the action), and the true reward is also state-only, and the dynamics satisfy a "decomposability condition" (all states linked through successor relationships), then Fu et al. prove $g^*(s) = r(s)+\text{const}$ and $h^*(s) = V^*(s)+\text{const}$. This is exact recovery, but under two strong requirements: (i) $r(s,a) = r(s)$ must hold, and (ii) the dynamics must satisfy decomposability. In many economic applications, rewards naturally depend on actions (e.g., paying vs. waiting), violating (i).

5.2 The LSW Anchor Identification Strategy

Lee et al. (2026), henceforth LSW, do not rely on the state-only restriction or the generic cross-environment argument. Instead, they impose two anchors:
(A1) Exit payoff anchor: $g(s, \text{exit}) = 0$ for all $s \in \mathcal{S}$.
(A2) Absorbing-state anchor: $h(s_{\text{abs}}) = 0$, where s_{abs} is the absorbing state reached after exit.
The exit action deterministically transitions to s_{abs}: $P(s_{\text{abs}} \mid s, \text{exit}) = 1$. These anchors eliminate the shaping ambiguity entirely.

Lemma 1 (General non-identifiability; LSW Appendix W4)**.** *If two pairs (g,h) and $(\bar{g},\bar{h})$ both produce the same AIRL score f for all (s,a,s'), then there exists a function $\Phi : \mathcal{S} \to \mathbb{R}$ such that*

$$\bar{h}(s) = h(s) + \Phi(s), \qquad \bar{g}(s,a) = g(s,a) + \Phi(s) - \beta\Phi(s').$$

Proof. The equality $g(s,a) + \beta h(s') - h(s) = \bar{g}(s,a) + \beta\bar{h}(s') - \bar{h}(s)$ for all (s,a,s') rearranges to $\bar{g}(s,a) - g(s,a) = [h(s) - \bar{h}(s)] - \beta[h(s') - \bar{h}(s')]$. Defining $\Phi = \bar{h} - h$ gives the result. □ □

Theorem 2 (Identification with anchors; LSW Theorem 1)**.** *Under anchors (A1)–(A2), if (g,h) and $(\bar{g},\bar{h})$ both realize the same f, then $g = \bar{g}$ and $h = \bar{h}$.*

Proof. By Lemma 1, there exists Φ with $\bar{g}(s,a) = g(s,a) + \Phi(s) - \beta\Phi(s')$ and $\bar{h}(s) = h(s) + \Phi(s)$. Apply (A1) at $a = \text{exit}$: $0 = \bar{g}(s, \text{exit}) = 0 + \Phi(s) - \beta\Phi(s_{\text{abs}})$, so $\Phi(s) = \beta\Phi(s_{\text{abs}})$ for all s. Apply (A2): $0 = \bar{h}(s_{\text{abs}}) = 0 + \Phi(s_{\text{abs}})$, so $\Phi(s_{\text{abs}}) = 0$. Therefore $\Phi \equiv 0$. □ □

Theorem 3 (Recovery of true utilities; LSW Theorem 2)**.** *Suppose the true Bellman pair (r^*, V^*) satisfies $A^*(s,a) = r^*(s,a) + \beta\, \mathbb{E}_P[V^*(s') \mid s,a] - V^*(s)$ with $r^*(s, \text{exit}) = 0$ and $V^*(s_{\text{abs}}) = 0$. Let (g,h) be any AIRL pair realizing the same advantage under the same anchors. Then $g = r^*$ and $h = V^*$.*

Proof. Subtract the two representations: $(g - r^*)(s,a) + \beta\, \mathbb{E}_P[(h - V^*)(s') \mid s,a] - (h - V^*)(s) = 0$. Define $\Phi = h - V^*$. At $a = \text{exit}$: $0 = \Phi(s) - \beta\Phi(s_{\text{abs}})$. Anchor (A2) applied to both h and V^* gives $\Phi(s_{\text{abs}}) = 0$, hence $\Phi \equiv 0$. □ □

Remark 4 (Deterministic transitions)**.** *In the LSW application, state transitions are deterministic conditional on the action and series structure. This means* $\mathbb{E}_P[h(s') \mid s, a] = h(s')$ *with no averaging, which makes the shaping decomposition* (7) *exact at each realized transition rather than holding only in expectation. The anchor proofs go through identically (and more cleanly) in this case.*

Remark 5 (Comparison to standard DDC normalization)**.** *In conventional DDC models, normalizing the outside option payoff to zero suffices because the Bellman equation is solved analytically under known transitions, pinning down continuation values. In AIRL, both* g *and* h *are learned jointly by gradient descent rather than solved for. Normalizing exit alone leaves* h *undetermined. The absorbing-state anchor (A2) fixes this remaining degree of freedom.*

Remark 6 (Relation to Kang et al. (2025))**.** *Kang et al. develop an ERM-based offline IRL/DDC framework that also uses anchor actions to achieve identification without requiring explicit transition probability estimation. Their approach minimizes Bellman residual error evaluated at anchor state-action pairs, bypassing the need for the adversarial loop entirely. The anchor action principle is the same–fixing the reward of a known action pins down levels–but the estimation strategy differs: Kang et al. solve a single (non-adversarial) ERM problem, while LSW use the anchors within the AIRL discriminator structure. The adversarial approach has the advantage of not requiring even a first-stage CCP estimator, at the cost of the instabilities inherent in GAN training.*

6 Approach 3: Inverse Soft-Q Learning (IQ-Learn)

Garg et al. (2021) observe that under the soft Bellman equation, the reward can be expressed as a function of Q alone:

$$r(s, a) = Q(s, a) - \beta \sum_{s'} P(s' \mid s, a)\, V_Q(s'), \tag{10}$$

where $V_Q(s) = \log \sum_{a'} \exp Q(s, a')$. Call this the *inverse Bellman operator*: $r_Q \equiv \mathcal{T}^{-1}[Q]$.

IQ-Learn estimates a single Q-function by solving:

$$\max_Q \; \mathbb{E}_{(s,a)\sim\rho^*}\big[\phi\big(r_Q(s, a)\big)\big] - (1 - \beta)\, \mathbb{E}_{s_0\sim\mu_0}\big[V_Q(s_0)\big], \tag{11}$$

where ρ^* is the expert's occupancy measure, μ_0 is the initial state distribution, and $\phi : \mathbb{R} \to \mathbb{R}$ is a concave function determining the divergence measure (e.g., $\phi(x) = x - x^2/2$ for χ^2).

What it recovers. At the optimum, $\hat{r}_{\mathrm{IQ}}(s, a) = Q^*(s, a) - \beta \sum_{s'} P(s' \mid s, a) V_{Q^*}(s')$ separates current payoffs from continuation values via the Bellman constraint. Unlike the reduced-form approach, IQ-Learn uses transition data, so the estimated reward is not merely the advantage.

Computational cost. Single optimization over Q-function parameters. No adversarial loop, hence more stable training than AIRL. Moderate cost (between reduced-form and AIRL).

7 Approach 4: GLADIUS

GLADIUS (Kang et al., 2025) parameterizes both $Q(s, a)$ and $\mathbb{E}[V(s') \mid s, a]$ with neural networks (MLPs) and trains them jointly on observed (s, a, s') transitions. The Q-network is trained via a negative log-likelihood loss (the same logit objective as the reduced-form approach) plus a Bellman consistency penalty that pushes $\mathbb{E}[V(s') \mid s, a]$ toward $V(s') = \log \sum_{a'} \exp Q(s', a')$ evaluated at observed next states.

The reward is extracted via the same inverse Bellman operator as IQ-Learn:

$$\hat{r}_{\text{GLADIUS}}(s, a) = Q(s, a) - \beta\, \hat{\mathbb{E}}[V(s') \mid s, a],$$

where $\hat{\mathbb{E}}[V(s') \mid s, a]$ comes from the learned EV-network rather than from the known transition matrix.

At the population level, GLADIUS recovers r^* exactly (same theoretical guarantee as IQ-Learn). In finite samples, the neural network architecture provides an important practical advantage over tabular methods: it shares parameters across states through the hidden layers, which regularizes the Q-function implicitly and reduces the effective dimensionality of the optimization. Where a tabular Q has $|\mathcal{S}| \times |\mathcal{A}|$ free parameters (543 in our simulation), a two-layer MLP with 64 hidden units has far fewer effective degrees of freedom.

Computational cost. Neural network training via mini-batch SGD. More expensive per iteration than the tabular methods but better-conditioned. The alternating update scheme (Q-network on odd batches, EV-network on even batches) stabilizes training.

8 When Do the Methods Diverge?

8.1 Taxonomy of Counterfactuals

A counterfactual modifies one or more primitives of the MDP and asks how optimal behavior changes. We distinguish three types.

Definition 7 (Type I: State-value extrapolation)**.** *The transition kernel P and reward r are unchanged. Only the realized state values shift (e.g., wait times decrease). The counterfactual policy is $\tilde{\pi}(a \mid \tilde{s}) = \exp(Q^*(\tilde{s}, a) - V^*(\tilde{s}))$, which requires only evaluating the already-estimated Q-function at new inputs. No Bellman equation is re-solved.*

Definition 8 (Type II: Change in the MDP environment)**.** *The reward r is unchanged, but the environment shifts: the transition kernel $P \to \tilde{P}$, the action-set structure changes (e.g., some episodes become paid-only), or access rules are modified by segment. The counterfactual Q-function $\tilde{Q}$ must satisfy a new Bellman equation $\tilde{Q}(s, a) = r(s, a) + \beta \sum_{s'} \tilde{P}(s' \mid s, a)\tilde{V}(s')$. This requires the structural reward $r(s, a)$ separated from the old continuation values.*

Definition 9 (Type III: Change in reward levels)**.** *The functional form of r itself changes (e.g., a tax on certain actions), or the discount factor shifts. This requires knowledge of r in levels, not just differences.*

8.2 Equivalence Under Type I

Proposition 10. *Under a Type I counterfactual, all three methods yield identical counterfactual choice probabilities.*

Proof. All three recover scores whose within-state action differences match Q^*. This includes $Q^*(s, a) + c(s)$ (reduced-form), $A^*(s, a) = Q^*(s, a) - V^*(s)$ (AIRL), and Q^* itself (IQ-Learn with anchors). Denote any such estimate as $Q^*(s, a) + c(s)$. For any $\tilde{s}$:

$$\tilde{\pi}(a \mid \tilde{s}) = \frac{\exp(Q^*(\tilde{s}, a) + c(\tilde{s}))}{\sum_{a'} \exp(Q^*(\tilde{s}, a') + c(\tilde{s}))} = \frac{\exp(Q^*(\tilde{s}, a))}{\sum_{a'} \exp(Q^*(\tilde{s}, a'))}.$$

The shift $c(\tilde{s})$ cancels. □ □

8.3 Divergence Under Type II

Proposition 11. *Under a Type II counterfactual, the reduced-form approach produces invalid predictions. AIRL with the LSW anchors and IQ-Learn can produce valid predictions, provided the structural reward is correctly recovered.*

The argument is as follows. The reduced-form Q-estimate is $Q^*(s,a) + c(s)$, which bundles $r(s,a)$ with $\beta \sum_{s'} P(s' \mid s,a)V^*(s')$. Under the new environment $\tilde{P}$, continuation values change, but the reduced-form estimate has no way to disentangle r from the old continuation structure. Re-solving the Bellman equation with A^* as the "reward" produces incorrect counterfactual CCPs because A^* already incorporates the original P.

AIRL with LSW's anchors recovers $g^* = r^*$ exactly (Theorem 3), so the structural reward is available for re-solving under $\tilde{P}$. Similarly, IQ-Learn's inverse Bellman operator separates r from continuation values by enforcing Bellman consistency.

8.4 Connection to KSS Partial Identification

Kalouptsidi et al. (2021) establish that in DDC models:

- Counterfactual behavior is identified under pre-specified additive payoff changes–i.e., $\tilde{r}(s,a) = r(s,a) + \Delta(s,a)$ where Δ is known. Type I counterfactuals fall in this class.
- Counterfactual behavior is generally not identified under changes to the transition kernel $P \to \tilde{P}$ without additional restrictions.

The LSW anchor strategy provides exactly the "additional restrictions" that move their counterfactuals from the non-identified to the identified case: by pinning down r in levels (via exit normalization) and V in levels (via absorbing-state normalization), they collapse the identified set to a singleton.

9 Application to Lee, Sudhir & Wang (2026)

LSW study sequential reading behavior on a serialized fiction platform. Readers decide at each episode whether to pay for immediate access, wait for free access, or exit. Each book defines a deterministic content trajectory. The state includes text features (32-dimensional autoencoder embeddings), reading progress, wait-time, and prior payment behavior. Four latent segments are estimated; two (Pay & Read, Wait & Read) dominate engagement.

9.1 Their Identification Strategy

LSW impose anchors (A1)–(A2): exit payoff is zero; the absorbing post-exit state has zero continuation value. By Theorems 2–3, this pins down both $g_\theta(s,a,z) = r^*(s,a,z)$ and $h_\phi(s,z) = V^*(s,z)$ for each segment z. The state-only restriction of Fu et al. is not needed. Action-dependent utilities are identified.

The deterministic transition structure further strengthens the result: no averaging over next states is needed in the decomposition (7), so the shaping term $\beta h(s') - h(s)$ is evaluated at the exact realized transition.

9.2 Classification of Their Counterfactuals

LSW run three classes of counterfactuals. We classify each.

Validation exercise (Section 6.3): Type I. Wait times are reduced for a held-out set of series. "No parameters are re-estimated; only the wait-time variable in the state space is modified" (LSW, p. 27). This feeds new state values into the fixed policy. By Proposition 10, a reduced-form logit would produce identical predictions.

WFF removal and delay variation (Sections 7.1–7.2): Type II. These counterfactuals change the environment: some episodes switch from WFF to paid-only access (altering the action-set structure at each episode), and wait times are varied differentially by segment (changing the MDP that each type faces). Trajectories are re-simulated through the MDP using the recovered $g_\theta(s, a, z)$ as the structural reward. A reduced-form Q-estimate cannot support these counterfactuals because it cannot separate reward from continuation values under the original environment.

Content-based pricing (Section 7.3): Type II. Episodes above a purchase-margin threshold become paid-only while others remain WFF. This again changes the action-set structure episode-by-episode, requiring re-simulation with the structural reward. The content-aware paywall placement exploits the fact that $g_\theta(s, a, z)$ captures how content enters immediate utility, which would be confounded with continuation effects in a reduced-form estimate.

9.3 What the Reduced-Form Approach Could and Could Not Replicate

A reduced-form logit with the same neural network architecture would:

- Match in-sample fit identically (same likelihood).
- Replicate the validation exercise (Type I) identically.
- Fail on all Section 7 counterfactuals (Type II), because these change the MDP environment and require the structural reward.

The validation exercise is, by design, easy for any method that recovers the advantage function–a reduced-form logit would pass it identically. The policy counterfactuals that constitute the paper's main contribution, however, require the structural decomposition that AIRL with anchors provides.

10 Simulation Study

10.1 Design

We construct a tabular MDP that mimics the LSW book-reading environment. The state encodes episode position ($p \in \{0, \ldots, 14\}$), content quality ($c \in \{\text{low, med, high}\}$), and wait-time level ($w \in \{1, 6, 12, 24\}$ hours), giving $15 \times 3 \times 4 = 180$ regular states plus one absorbing state. Actions are {buy, wait, exit}. Transitions are deterministic: buy and wait advance to the next episode; exit leads to the absorbing state. The true reward is

$$\begin{aligned} r^*(s, \text{buy}) &= u(c) - p_{\text{price}}, \\ r^*(s, \text{wait}) &= u(c) - \delta \cdot w, \\ r^*(s, \text{exit}) &= 0, \end{aligned}$$

where $u(\text{low}) = 0.5$, $u(\text{med}) = 1.5$, $u(\text{high}) = 3.0$, $p_{\text{price}} = 2.0$, $\delta = 0.08$, and $\beta = 0.95$. Five methods are compared at the population level (no estimation noise): Oracle (r^*), AIRL+anchors ($g^* = r^*$ by Theorem 3), IQ-Learn ($r_Q = Q^* - \beta V^*(s')$ via inverse Bellman), AIRL-no-anchors ($g = r^* + \alpha V^*(s) - \alpha\beta V^*(s')$ with $\alpha = 0.5$), and Reduced-form ($A^* = \log \pi^*$).

10.2 Analysis A: Reward Recovery

Type I counterfactual (changing wait-time from 24h to 1h): all five methods produce CCP errors below 10^{-6}, confirming Proposition 10 that the action-independent shift $c(s)$ cancels in the softmax.

Method	MSE vs. r^*	Corr. with r^*
Oracle	0.000	1.000
AIRL+anchors	0.000	1.000
IQ-Learn/GLADIUS	0.000	1.000
AIRL-no-anchors	10.005	0.440
Reduced-form	40.020	0.138

Table 1: Reward recovery at population convergence. Oracle, AIRL+anchors, IQ-Learn, and GLADIUS recover r^* exactly. AIRL-no-anchors recovers a shaped perturbation. Reduced-form recovers the advantage A^*, which has low correlation with r^* because it bundles immediate payoff with continuation values.

10.3 Analysis B: Type II Error vs. Shift Magnitude

The counterfactual changes the transition kernel so that buying advances k episodes (instead of 1). The Bellman equation must be re-solved under each method's recovered reward.

k	Oracle / AIRL+a / IQ / GLADIUS	AIRL-no-a	Reduced-form
1	0.000	0.000	0.000
2	0.000	0.025	0.082
3	0.000	0.017	0.094
5	0.000	0.015	0.123
7	0.000	0.016	0.129
10	0.000	0.014	0.139

Table 2: Mean absolute CCP error vs. oracle under Type II counterfactuals. Oracle, AIRL+anchors, IQ-Learn, and GLADIUS are exact at every k. Reduced-form error grows monotonically with the magnitude of the transition shift.

10.4 Analysis C: Shaping Magnitude Sweep

To show that counterfactual validity depends on how far the recovered reward is from r^*, we construct shaped rewards $\tilde{r} = r^* + \alpha\,\delta(s) - \alpha\beta\,\delta(s')$ where $\delta = V^*$ and $\alpha \in [0, 1]$. At $\alpha = 0$, the reward is exact; at $\alpha = 1$, it is maximally shaped. We evaluate the Type II counterfactual with $k = 3$.

10.5 Analysis D: Finite-Sample Convergence

At the population level, anchor-constrained methods recover r^* exactly (Analysis A). In finite samples, an important practical question arises: does the structural approach beat the simpler reduced-form alternative, or does the optimization difficulty erase the theoretical advantage?

α	Type II CF error	Corr. with r^*
0.00	0.000	1.000
0.05	0.002	0.974
0.10	0.004	0.908
0.20	0.008	0.746
0.30	0.011	0.610
0.50	0.017	0.440
0.70	0.022	0.347
1.00	0.027	0.270

Table 3: Counterfactual error is monotone in shaping magnitude α. Without anchors, which value of α the optimizer converges to is indeterminate–the counterfactual accuracy is a lottery over the identified set.

The standard IQ-Learn χ^2 objective over 543 tabular parameters has multiple local optima in finite samples. Its global finite-sample optimum differs from the population optimum (the true Q^*), so even with L-BFGS-B convergence, it produces worse counterfactual errors than the reduced-form baseline. This is a known fragility of the IQ-Learn χ^2 objective in offline tabular settings.

We resolve this by replacing the χ^2 objective with an anchor-constrained MLE: the standard logit likelihood (convex, fast convergence) plus soft anchor penalties on the implied reward $r_Q(s,a) = Q(s,a) - \beta \sum_{s'} P(s' \mid s,a) V_Q(s')$. The penalty terms $\|r_Q(\cdot, \text{exit})\|^2$ and $\|r_Q(s_{\text{abs}}, \cdot)\|^2$ pin down Q-function levels without distorting within-state action differences. This is conceptually close to the Kang et al. (2025) ERM approach, where anchor actions achieve identification within a single optimization.

We also compare against GLADIUS, which parameterizes Q and $\mathbb{E}[V(s')]$ with neural networks. The MLP architecture shares parameters across states, providing implicit regularization that the tabular methods lack.

N	Reduced-form	Anchor-MLE	GLADIUS
500	0.264	0.255	0.106
2,000	0.262	0.255	0.123
10,000	0.262	0.257	0.099

Table 4: Finite-sample Type II counterfactual error (2 seeds, $k = 3$). The reduced-form logit converges to the wrong target (A^*), plateauing at 0.262. The anchor-constrained tabular MLE improves modestly to 0.255. GLADIUS achieves substantially lower error (0.10) because the neural network architecture shares parameters across states, providing implicit regularization that the 543-parameter tabular Q-function lacks. The estimation framework matters in practice, not just the identification strategy.

10.6 Analysis E: Anchor Misspecification Robustness

What if the true exit payoff is $\varepsilon > 0$ but the anchors assume it is zero? We vary ε and track the effect on reward recovery and Type II counterfactual accuracy.

ε	Corr. with r^*	Type II CF error	In-sample π shift
0.0	1.000	0.000	0.000
0.1	0.999	0.000	0.000
0.3	0.991	0.001	0.000
0.5	0.976	0.002	0.000
1.0	0.909	0.004	0.000
2.0	0.729	0.008	0.000

Table 5: Anchor misspecification robustness. The Type II error grows smoothly with ε but remains small. In-sample CCPs are unaffected (the policy is still correctly recovered). The identification is not knife-edge: moderate misspecification of the exit payoff produces moderate, not catastrophic, errors.

10.7 Analysis F: Content-Based Paywall Counterfactual

This counterfactual mirrors Section 7.3 of LSW. High-content episodes (quality level 2, at positions 0, 1, 7, 8, 9, 13, 14) become paid-only: the "wait" action is blocked, forcing buy-or-exit. This changes the action-set structure episode-by-episode–a genuine Type II shift.

Method	CCP error vs. oracle
Oracle	0.000
AIRL+anchors	0.000
IQ-Learn	0.000
AIRL-no-anchors	0.009
Reduced-form	0.185

Table 6: Content-based paywall counterfactual. The reduced-form approach fails badly (18.5pp mean CCP error) because it cannot separate how content enters immediate utility vs. continuation value. Structural methods are exact.

The reduced-form error of 18.5 percentage points is economically significant: at the mean exit rate of 2.8 percent, this represents a several-fold misprediction of dropout behavior at paywall episodes. This is exactly the setting where a platform would make incorrect pricing decisions by relying on a reduced-form model.

11 Summary

The key insight is that AIRL with anchors achieves what generic AIRL cannot guarantee: exact recovery of the structural reward, collapsing the identified set to a singleton. For the LSW application, the exit and absorbing-state normalizations–combined with deterministic transitions–make $g^* = r^*$ and $h^* = V^*$ provably, not approximately. This is what licenses the Type II counterfactuals in their paper and what would be lost under a simpler reduced-form approach.

References

Cao, H., Cohen, S., and Szpruch, L. (2021). "Identifiability in Inverse Reinforcement Learning." NeurIPS 34, 12362–12373.

	Reduced-form Q	AIRL (generic)	AIRL (anchors)	IQ-Learn / GLADIUS
Optimization	Single MLE	Minimax (GAN)	Minimax (GAN)	Single max / SGD
Uses $P(s' \mid s, a)$?	No	Implicit (sim.)	Implicit (sim.)	Yes / Empirical
Recovers	$A^*(s, a)$	r + shaping	$r^*(s, a)$	$r^*(s, a)$
Type I CF	✓	✓	✓	✓
Type II CF	×	Only if $\delta \equiv 0$	✓	✓
Type III CF	×	×	Requires levels	Requires levels
Stability	Convex	Adversarial	Adversarial	Stable

Table 7: Comparison of four approaches to reward recovery.

Fu, J., Luo, K., and Levine, S. (2018). "Learning Robust Rewards with Adversarial Inverse Reinforcement Learning." ICLR.

Garg, D., Chakraborty, S., Cundy, C., Song, J., and Ermon, S. (2021). "IQ-Learn: Inverse soft-Q Learning for Imitation." NeurIPS.

Ho, J. and Ermon, S. (2016). "Generative Adversarial Imitation Learning." NeurIPS.

Hotz, V. J. and Miller, R. A. (1993). "Conditional Choice Probabilities and the Estimation of Dynamic Models." Review of Economic Studies, 60(3), 497–529.

Kalouptsidi, M., Scott, P. T., and Souza-Rodrigues, E. (2021). "Identification of Counterfactuals in Dynamic Discrete Choice Models." Quantitative Economics, 12(2), 351–403.

Kang, E. H., Yoganarasimhan, H., and Jain, L. (2025). "An Empirical Risk Minimization Approach for Offline Inverse RL and Dynamic Discrete Choice Models." arXiv:2502.14131.

Lee, P., Sudhir, K., and Wang, T. (2026). "Consumer Engagement with Sequential Content: A Content-Aware Dynamic Choice Model." Working paper, Yale University.

Magnac, T. and Thesmar, D. (2002). "Identifying Dynamic Discrete Decision Processes." Econometrica, 70(2), 801–816.

Ng, A. Y., Haley, D., and Russell, S. J. (1999). "Policy Invariance Under Reward Transformations." ICML.

Rust, J. (1987). "Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher." Econometrica, 55(5), 999–1033.

Ziebart, B. D. (2010). Modeling Purposeful Adaptive Behavior with the Principle of Maximum Causal Entropy. PhD thesis, Carnegie Mellon University.